Mohamed Kamel

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Education

Virginia Commonwealth University (VCU) — *Richmond, VA, United States*

PH.D. IN COMPUTER SCIENCE — *2024 – Present*

- GPA: 4.0/4.0

Nile University (NU) — *Cairo, Egypt*

M.SC. IN INFORMATICS — *2022 – 2024*

- GPA: 3.97/4.0

Ahram Canadian University (ACU) — *Cairo, Egypt*

B.SC. IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY — *2018 – 2022*

- GPA: 3.9/4.0, Second of the class

Skills

Programming	Python, C++, Java, JavaScript | Data Structures & Algorithms
ML Frameworks	PyTorch, TensorFlow, CUDA
ML & Generative AI	Deep Learning, Reinforcement Learning, Generative AI, Transformers, Diffusion Models, Rectified Flow, Flow Matching
Signal Processing	Signal Separation, Biomedical Signal Processing, Time Series Analysis, RF Signal Processing, Semantic Communication
Vision & Multimodal	Computer Vision, Medical Imaging, U-Net, ViT, Semantic Segmentation, Self-Supervised Learning, Multimodal Learning
Behavioral AI	Human Activity Recognition, Behavioral Sensing, Smart-Home Sensing, Digital Health
LLMs	Prompt Engineering, LoRA, NER, LLM Poisoning
Software & Backend	FastAPI, Docker, REST APIs, Microservices, MongoDB, Firebase
Tools & Cloud	Git, Linux, AWS, Azure
Languages	Arabic: Native | English: C1 | German: A1

Publications

BeatFlow-ECG: Rectified Flow for ECG Reconstruction from Indirect Wearable Signals — *ICLR*

CONFERENCE PUBLICATION — *Under Review, 2027* | *View Paper*

TrustFlow-ECG: Reliability-Aware ECG Restoration with Calibrated Risk Estimation — *ICASSP*

CONFERENCE PUBLICATION — *Under Review, 2026*

FlowSep-RF: Rectified Flow Matching for Single-Channel Radio-Frequency Signal Separation — *ICASSP*

CONFERENCE PUBLICATION — *Under Review, 2026* | *View Paper*

Communication-Aware Rectified Flow Matching for Robust Single-Channel RF Source Separation — *IEEE Transactions on Wireless Communications (TWC)*

JOURNAL PUBLICATION — *Under Review, 2026*

SemTCP: Semantic-Aware Scheduling and Selective Recovery for Timely Task-Oriented Communication — *ICC*

CONFERENCE PUBLICATION — *Under Review, 2026*

Two-Stage User Activity Detection and Localization for Grant-Free Random Access in Cell-Free Massive MIMO — *ICC*

CONFERENCE PUBLICATION — *Under Review, 2026*

ProSeD: Source-Side Semantic Decoys Against Unauthorized Data Harvesting
Journal Publication
Knowledge-Based Systems (KBS)
Under Review, 2026

CondFlow-ECG: Conditional Rectified Flow for Morphology-Aware ECG Denoising
Conference Publication
ICMLA
Accepted, 2026

Development of an AI-Enabled Remote Loneliness Assessment System for Older Adults Using Behavioral and Speech Data from IoT Sensors and Smart Speakers
Journal Publication
Assessment (SAGE), Special Issue on AI in Psychological Assessment [View Paper]
2026

Multimodal Activity Episode Inference from Smart-Home Signals for Behavioral Health Monitoring in Older Adults
Conference Publication
MLSP
Under Review, 2026

URGE: Urgency-Aware Energy-Efficient UAV Mobility via Hybrid Deep Reinforcement Learning
Conference Publication
International Conference on Smart Mobility (SM) [View Paper]
2026

Attention-Driven Reinforcement Learning to Optimize Packet Retransmission in Semantic Communication
Conference Publication
IEEE GLOBECOM [View Paper]
2025

SkinFormer: Robust Vision Transformer for Automatic Skin Disease Identification
Conference Publication
JAC-ECC [View Paper]
2022

Academic Service

IET Communications journal
Reviewer
2026

International Conference on Machine Learning and Applications (ICMLA)
Reviewer
2026

Assessment journal
Reviewer
2026

International Conference on Smart Mobility (SM)
Reviewer
2026

Research Talks

From Wearable Signals to Clinical ECG: Toward Continuous Cardiac Monitoring
Invited Speaker
Department of Computer Science Seminar, Virginia Commonwealth University
March 20, 2026

Honors, Awards & Certifications

Badge issued by Virginia Commonwealth University (VCU)
Leadership Foundations for Graduate Students
Credly
Oct 2025

Semi-Finalist, Dell Technologies Envision the Future
Online

Graduation Project Competition (Middle East, Russia, Africa, Turkey
2022

Certificate, Software Design and Architecture Specialization University of Alberta
Online

Covered: Object-Oriented Design, Design Patterns, Software Architecture, Service-Oriented Architecture
2021

First Place, ACU Problem Solving Competition
Leaderboard

Hackerearth College Challenge
2020

Academic Excellence Award
Ahram Canadian University

Awarded Multiple Times for Top Academic Performance
2019 – 2022

Experience

VCU + Bank of America – Special Olympics
United States

Software Engineer Intern (Part-time)
May 2025 – Aug 2025

- lead the backend development of a software application supporting Special Olympics.
- Involved in software design, API development, testing, and integration with frontend components.
- Tech Used: Python, FastAPI, MongoDB, Azure, Figma, React, TypeScript, HTML, CSS, JavaScript, Git

Virginia Commonwealth University (VCU)
United States

Graduate Assistant
Jan. 2024 – Present

- **HomePAL:** Contributing to the NIH-funded AI-enabled health monitoring project (**NIH R01AG072611**, Project Link).
  - Working on sensor setup, data collection scripts, signal analysis, and AI-based behavioral anomaly detection.
- **ECG-Based ML for EP Guidance:** Working on an ECG-based machine learning project for electrophysiology procedures, with responsibility for the technical implementation and model development.
  - Developing automated ECG measurement methods for pacing analysis, including left ventricular activation time, QRS duration, and R-wave peak time.
- **HRV Biofeedback:** Contributing to the development of a wearable-sensor system for real-time ECG, HRV, and respiration monitoring and biofeedback during stationary cycling.
  - Working on integrating ECG, respiration, and wrist-based PPG/IMU sensors for real-time data collection and analysis.
- **FoodTrack:** Contributing to the SARE-funded food waste tracking and diversion project (**SARE FLW24-004**, Project Link).
- **Teaching Assistant:** Held weekly office hours, graded assignments, and provided academic support for undergraduate CS courses.

Nile University (NU)
Egypt

Teaching Assistant (Full-time)
Feb. 2023 – Jan. 2024

- Assisted in teaching AI, CS major, and programming courses.
- Provided lab support and one-on-one mentoring for students.

Ahram Canadian University (ACU)
Egypt

Teaching Assistant (Full-time)
Oct. 2022 – Feb. 2023

- Taught AI-related and core programming subjects.
- Supported student learning and practical lab implementation.

LÀCRIMA Dairy Industrial Ltd
Remote - Jordan

Software Developer (Contract)
Jan. 2022 – Jul. 2023

- Developed an internal desktop system for electronic file archiving and communication (Demo Video).
- Designed customer-facing and employee mobile applications to improve service interaction.(Android | iOS)
- Tech Used: Java, JavaFX, AWS, MySQL, Flutter, Firebase, Kotlin

Order GO
Remote - United Kingdom

Full Stack Developer (Full-time)
Apr. 2021 – Oct. 2021

- Built the company's public website and improved digital presence.
- Developed the ROS App for streamlining kitchen order management.
- Tech Used: Java, JavaFX, Kotlin, HTML, CSS, JavaScript, Bootstrap, PHP, Firebase, Git

Upwork
Freelance

Top-Rated Freelancer
Apr. 2019 – Oct. 2023

- Completed 65+ freelance software projects with a 98% Job Success Score.
- Delivered AI, web, mobile, and desktop solutions for international clients.

Volunteering Activities

Ahram Canadian University (ACU) *Egypt*

Volunteer Undergraduate Teaching Assistant *2018 – 2022*

- Supported courses in Linear Algebra, Programming, Event-Driven Programming, and Data Structures; created educational videos, with one reaching nearly 60,000 views (Video Link).

EXCEPTION Student Committee *Egypt*

Member and Academic Instructor *2018 – 2022*

- Conducted workshops in CS fundamentals, programming, Java (OOP), and Web development.

ACU CPC Community (ACM) *Egypt*

Mentor *2020 – 2022*

- Mentored peers in programming competitions and problem solving.

ACU DSC Community (Google Student Developer) *Egypt*

Member *2020 – 2021*

- Participated in technical workshops and peer-learning activities.

Selected Projects

Navigating the Retinal Maze: IR Imaging for Retinal Exploration *2024*

- Developed CrackRAVIR, a deep learning model for segmentation of retinal arteries and veins using IR imaging.
- Utilized Residual Attention UNET to improve segmentation accuracy and vascular width estimation.
- Improved glaucoma and AMD detection through detailed vasculature analysis.
- Technologies: Deep Learning, Residual Attention UNET, Semantic Segmentation, Python, IR Imaging.

Graph Neural Networks for Movie Recommendation Systems *2024*

- Developed a GraphSAGE-based recommendation system for movies using bipartite graphs of users/movies.
- Addressed cold start and scalability issues, compared with collaborative filtering and LightGCN.
- Technologies: Python, GraphSAGE, PyTorch Geometric, Collaborative Filtering, LightGCN.

Evidence Detection in Cloud Forensics *2023*

- Designed a system to detect evidence of cyber threats and anomalies in cloud environments.
- Utilized Gradient Boosting Machines, Sequential NN, and RNN to improve forensic accuracy and speed.
- Technologies: Python, GBM, RNN, Enhanced MLP.

AI Disease Diagnostic System Using Deep Learning and Image Processing *2022*

- Developed an application for early diagnosis of diseases using 50,000+ images from Fitzpatrick17k, SD198, and ISIC 2019 datasets.
- Achieved 85% accuracy using LAMB optimizer and advanced image augmentation techniques.
- Built with clean architecture and MVI framework.
- Technologies: Machine Learning, Deep Learning, Python, Kotlin, Vision Transformers, Neural Networks, MVI.
- Demo Preview | Paper

Self-Driving Car with Object Detection (CNN, RL, YOLOv3) *2022*

- Developed a self-driving car model capable of detecting traffic signals and adjusting speed dynamically.
- Technologies: Computer Vision, Carla Simulator, CNNs, Reinforcement Learning, YOLOv3, Python.

Extending Fastpitch with Emotional Awareness *2021*

- Enhanced Fastpitch for emotional speech output using a modified Conformer architecture.
- Trained with emotional labels and perceptual loss for expressive speech synthesis.
- Technologies: Python, Conformer, Fastpitch, DeepVoice3, Mel Spectrogram.

Early-Stage Diabetes Risk Prediction Classifier *2021*

- Developed a binary classifier for diabetes risk prediction using tabular data.
- Technologies: Python, FastAI, Machine Learning. GitHub Repo

References

Available upon request.